

The Effect of Public Awareness of Fatal Police Shootings on Mental Health-Related Emergency Medical Service Calls in New York City: A Distributed Lag Analysis with Heterogeneous Effects by Community District Demographics

Abraham Mahmud*

February 8, 2026

Abstract

This study examines the causal relationship between public awareness of fatal police shootings and mental health-related emergency medical service (EMS) calls in New York City. Using a distributed lag model with community district fixed effects, we analyze daily EMS call data from 2005-2025 and Twitter activity as a measure of public awareness from 2017-2020. We find a statistically significant increase in the mental health call share approximately 7 days after awareness spikes, with a coefficient of 0.00139 (SE: 0.00038, $p < 0.001$). The cumulative effect over 0-7 days is 0.00078 per one standard deviation increase in awareness, translating to approximately 33 additional mental health calls citywide per week during high-awareness periods. Critically, we document substantial heterogeneity in these effects across community districts based on their demographic composition. Districts with lower percentages of Black and Hispanic residents show stronger positive effects (coefficient: 0.00374 for highest % White quartile), while districts with higher percentages of Black residents exhibit negative or null effects at the 7-day lag (coefficient: -0.00035 for highest % Black quartile). Results are robust across multiple specifications including de-meaning, two-way fixed effects, quantile analysis, and difference-in-differences designs. The findings provide EMS with actionable intelligence: a 7-day predictive window enables proactive resource allocation, with district-specific targeting based on demographic composition. These results have important implications for understanding the public health consequences of police violence and for designing targeted mental health interventions and EMS operational protocols.

1 Introduction

Fatal police shootings represent a critical public health and social justice issue in the United States. Beyond the immediate loss of life, these events generate widespread public awareness through media coverage and social media, potentially affecting the mental health of communities

*MIT Undergraduate Research Opportunities Program (UROP) under Professor Justin Steil, Department of Urban Studies and Planning, Massachusetts Institute of Technology

far beyond those directly involved. Understanding the mental health consequences of exposure to information about police violence is essential for public health policy and resource allocation.

This study investigates whether increases in public awareness of fatal police shootings lead to changes in mental health-related emergency medical service (EMS) calls in New York City. We employ a distributed lag model to capture the temporal dynamics of this relationship, examining effects over a 28-day window following awareness spikes. Additionally, we examine how these effects vary across community districts based on their demographic and socioeconomic characteristics.

Our analysis contributes to the literature in several ways. First, we use high-frequency daily data to identify precise temporal patterns in mental health responses. Second, we leverage community district fixed effects to control for unobserved time-invariant characteristics. Third, we document substantial heterogeneity in effects across districts, suggesting that the mental health impacts of police violence awareness are not uniform across communities. This heterogeneity has important implications for understanding mechanisms and designing interventions.

2 Background and Literature

2.1 Police Violence and Mental Health

A growing body of research documents the mental health consequences of exposure to police violence. Studies have found associations between police killings and adverse mental health outcomes, including depression, anxiety, and post-traumatic stress disorder (3; 7). These effects extend beyond direct victims to include family members, witnesses, and community members who learn about incidents through media coverage.

The mechanisms through which awareness of police violence affects mental health are multifaceted. Direct exposure to violence can cause trauma, but vicarious exposure through media and social media can also trigger stress responses, particularly among individuals who identify with victims or who have experienced similar trauma (9). The racialized nature of police violence means that communities of color may be disproportionately affected, both through direct exposure and through the cumulative psychological toll of repeated exposure to violence against members of their community.

2.2 Social Media and Public Awareness

Social media platforms, particularly Twitter, have become central to how information about police violence is disseminated and discussed. Twitter activity provides a real-time measure of public attention to these events, capturing both media coverage and public discourse. Previous research has used Twitter data to measure public attention to various events, including police violence (8).

The use of social media data as a measure of awareness has several advantages. First, it provides daily-level granularity, allowing for precise temporal analysis. Second, it captures both traditional media coverage and organic public discussion. Third, it can be aggregated to create citywide measures that reflect overall levels of public attention.

2.3 Emergency Medical Services as a Mental Health Outcome

EMS calls represent an important measure of acute mental health needs. Mental health-related EMS calls include calls for emotional distress, suicide attempts, drug overdoses, and other mental health crises. These calls reflect individuals in acute distress who require immediate intervention, making them a particularly relevant outcome for understanding the immediate mental health consequences of traumatic events.

Using EMS data has several advantages. First, it provides objective, administrative data that is not subject to recall bias or selection into survey participation. Second, it captures severe cases that require emergency intervention, which may be more sensitive to acute stressors. Third, it is available at high frequency (daily) and fine geographic granularity (community district level), enabling precise temporal and spatial analysis.

3 Data

3.1 Emergency Medical Service Data

We use EMS incident dispatch data from New York City OpenData, covering the period from January 1, 2005 through August 31, 2025. The data include all EMS calls dispatched in New York City, with information on incident datetime, community district, final call type, and various indicators for call characteristics.

Mental Health Call Classification: We classify calls as mental health-related based on the following final call type codes:

- **EDP:** Emotionally Disturbed Person
- **ALTMEN, ALTMFC, ALTMFT:** Various types of mental health alerts
- **JUMPDN, JUMPUP, JUMPDC:** Suicide-related calls (jumping incidents)
- **OD, ODC:** Drug overdose calls
- **POISON:** Poisoning calls
- **DRUG:** Drug-related calls

Data Cleaning: We exclude the following types of calls to focus on routine emergency responses:

- Special events (e.g., parades, festivals)
- Standby calls (pre-positioned units)
- Transfer calls (inter-facility transfers)
- Cancelled calls
- Duplicate calls
- Calls with disposition code 87 (specific exclusion category)

Panel Construction: We aggregate the data to a community district $\times$ day panel, creating the following variables:

- **total_calls:** Total number of EMS calls in district i on day t
- **mh_calls:** Number of mental health-related calls
- **mh_share:** Mental health calls as a proportion of total calls ($mh_calls/total_calls$)

We restrict the sample to days with at least 5 total calls to ensure stable estimates of the mental health share. The final panel includes 485,113 district-day observations across 71 community districts.

3.2 Twitter Awareness Data

We measure public awareness of fatal police shootings using daily Twitter activity. The data come from two sources:

- Daily tweet counts for individual fatal police shooting incidents (2017-2018)
- Final daily tweet count aggregations (2018-2020)

We combine these datasets to create a continuous daily time series of citywide Twitter activity related to fatal police shootings. The raw tweet counts are z-scored (standardized) to create a measure of relative awareness, where values represent standard deviations from the mean. This standardization allows for interpretation in terms of typical variation in awareness levels.

The awareness measure is citywide (not district-specific), reflecting the fact that information about police shootings typically spreads across the entire city through media and social media. The z-scoring ensures that the measure is comparable across the study period and that coefficients represent effects per one standard deviation increase in awareness.

High Awareness Event Days: For threshold-based analyses, we define "high awareness event days" as calendar dates when the z-scored awareness measure exceeds specified thresholds (typically 1.0, 1.5, or 2.0 standard deviations above the mean). Importantly, these event days represent when *public awareness peaked*, which may be the same day as a police killing, or days later when news spreads and public attention builds. For example, while George Floyd was killed on May 25, 2020, the highest awareness event day was May 29, 2020 ($awareness_z = 20.43$), *representing when public attention and Twitter activity reached their peak. This distinction is important because*

3.3 Community District Demographics

To examine heterogeneous effects, we merge demographic data from the 2010 U.S. Census for each of New York City's 59 community districts. The data come from the NYC Department of City Planning's Community District Profiles and include:

- **Race/Ethnicity:** Percentage White (non-Hispanic), Black (non-Hispanic), Hispanic, and Asian (non-Hispanic)

- **Housing:** Total households, percentage owner-occupied, percentage renter-occupied

These demographic characteristics are time-invariant (measured at the 2010 Census) and are merged to all district-day observations in the panel. We use these variables to examine how effects vary by district demographics through interaction terms and stratified analysis.

3.4 Control Variables

We include several time-varying controls to account for confounding factors:

- **Day of week:** Fixed effects for each day of the week (Monday through Sunday)
- **Month $\times$ Year interactions:** Separate fixed effects for each month-year combination to control for seasonal patterns and long-term trends
- **Federal holidays:** Indicator for U.S. federal holidays
- **Community district fixed effects:** Time-invariant district characteristics

Notably, we do not include COVID-19 phase indicators, as these are perfectly collinear with the month $\times$ year fixed effects for 2020-2021.

4 Methods

4.1 Empirical Strategy

We employ a distributed lag model with community district fixed effects to estimate the effect of public awareness on mental health EMS calls. The baseline specification is:

$$mh_share_{it} = \alpha_i + \sum_{k=0}^{28} \beta_k \cdot awareness_{t-k} + X'_{it}\gamma + \varepsilon_{it} \quad (1)$$

where:

- mh_share_{it} is the mental health call share in district i on day t
- α_i are community district fixed effects
- $awareness_{t-k}$ is the z-scored Twitter awareness measure at lag k
- β_k are the lag-specific coefficients (the impulse response function)
- X_{it} includes day of week, month $\times$ year, and holiday indicators
- ε_{it} is the error term

We estimate the model using ordinary least squares (OLS) with standard errors clustered at the community district level to account for correlation of errors within districts over time.

4.2 Lag Specification

We examine lags at $k \in \{0, 1, 2, 3, 7, 14, 21, 28\}$ days. This specification allows us to capture:

- Immediate effects (lag 0)
- Short-term effects (lags 1-3 days)
- Medium-term effects (lag 7 days)
- Longer-term effects (lags 14, 21, 28 days)

The 7-day lag is of particular interest, as it may capture delayed mental health responses that develop over the course of a week following awareness spikes.

4.3 Robustness Analysis: Progressive Model Specification

To assess the robustness of our findings, we estimate a series of models that progressively add control variables:

1. **Model 1:** Awareness lags only
2. **Model 2:** + Day of week fixed effects
3. **Model 3:** + Month×Year fixed effects
4. **Model 4:** + Holiday indicator
5. **Model 5:** + Community District fixed effects

This approach allows us to examine how the coefficient of interest (lag 7) changes as we add controls, providing evidence on whether the effect is robust to different control strategies.

4.4 Awareness Threshold Analysis

As an alternative to the continuous distributed lag approach, we also analyze the effects using threshold-based indicators. We create binary indicators for "high awareness days" defined as days where awareness *exceeds specified thresholds* (1.0, 1.5, or 2.0 standard deviations). *This approach :*

- Captures extreme awareness events that may have proportionally larger effects
- Allows for clearer causal interpretation in difference-in-differences designs
- Provides policy-relevant thresholds for intervention timing

We estimate models with lagged high awareness indicators at lags 0, 1, 2, 3, 7, and 14 days, using the same control structure as the main distributed lag model.

4.5 Difference-in-Differences Analysis

We implement a difference-in-differences (DID) design using high awareness days ($\text{awareness}_z > 1.5SD$) as the treatment. The treatment group consists of days with high awareness, while the control group consists of days with low awareness.

$$mh_share_{it} = \alpha_i + \beta \cdot (\text{treated}_i \times \text{post}_t) + X'_{it}\gamma + \varepsilon_{it} \quad (2)$$

where treated_i indicates high awareness days and post_t indicates days after the event. We also estimate an event study specification with separate coefficients for each day relative to the event (from -14 to +14 days) to examine dynamic effects and test for pre-trends.

4.6 Heterogeneous Effects Analysis

To examine how effects vary by district demographics, we employ two complementary approaches:

Approach 1: Interaction Terms

We extend the baseline model to include interactions between awareness lags and district demographic characteristics:

$$mh_share_{it} = \alpha_i + \sum_{k \in K} \beta_k \cdot \text{awareness}_{t-k} + \sum_{k \in K} \sum_{d \in D} \gamma_{kd} \cdot (\text{awareness}_{t-k} \times \text{demo}_d^i) + X'_{it}\delta + \varepsilon_{it} \quad (3)$$

where $K = \{0, 1, 7, 14\}$ are key lags and D are demographic variables (% Black, % Hispanic, % White, % Asian). The interaction coefficients γ_{kd} capture how the effect of awareness varies by district demographics.

Approach 2: Stratified Analysis

We estimate separate models for each quartile of each demographic variable. This allows us to compare effects across districts with different demographic compositions without assuming linearity in the interaction effects.

4.7 Placebo Test

As a placebo test, we estimate the same model using total call volume as the outcome. If our measure of awareness is capturing a true mental health effect rather than a general increase in EMS utilization, we should see effects on mental health share but not on total calls.

4.8 Additional Robustness Checks

4.8.1 De-meaning by Day-of-Week and Community District

To address concerns that day-of-week patterns varying by district might drive our results, we implement a de-meaning procedure. We calculate the mean mental health call share for each weekday within each community district, then subtract this mean from each observation. This removes district-specific day-of-week patterns while preserving the variation of interest.

When we re-estimate our main model using the de-measured outcome variable, we find nearly identical results. The lag 7 coefficient is 0.001389 in the de-measured model compared to 0.001387

in the baseline model, a difference of only 0.14%. The cumulative effect over 0-7 days is 0.000772 in the de-meaned model compared to 0.000775 in the baseline, confirming that day-of-week patterns are not driving our findings.

4.8.2 Two-Way Fixed Effects

We also estimate models with both community district and date fixed effects to control for time-varying factors affecting all districts equally. Due to computational constraints with daily data (1,490 unique dates), we implement a weekly fixed effects approach as a computationally efficient alternative. The weekly fixed effects model shows a lag 7 coefficient of 0.001276 (SE: calculated from variance-covariance matrix), very similar to the one-way fixed effects estimate of 0.001387, confirming that time-varying confounders are well-controlled in our main specification.

4.8.3 Quantile-Based Analysis

To examine non-linear effects, we split awareness into quintiles and estimate separate models within each quantile. The top quintile (80-100%) shows a significant positive lag 7 coefficient of 0.001529 (SE: 0.000383), consistent with our main findings. This suggests that effects are concentrated during high-awareness periods, supporting the threshold-based approach.

4.8.4 Forward-Looking Lag Analysis

To address reverse causality concerns, we examine how awareness on day t affects mental health calls on future days ($t + 1$ through $t + 7$). We find that forward effects are smaller than backward effects: the forward lag 7 effect is 0.000457 (SE: 0.000265) compared to the backward lag 7 effect of 0.001230 (SE: 0.000351). This pattern supports our causal interpretation that awareness causes changes in mental health calls rather than the reverse.

5 Results

5.1 Main Effects

Table 1 presents the impulse response function coefficients from the distributed lag model with community district fixed effects (Model 5). The key finding is a statistically significant positive effect at the 7-day lag: $\hat{\beta}_7 = 0.00139$ (SE: 0.00038, $p < 0.001$).

Table 1: Impulse Response Function: Effect of Awareness on Mental Health Call Share

Lag (days)	Coefficient	Standard Error	95% Confidence Interval	Note: Standard errors
0	0.00050	0.00033	[-0.00015, 0.00115]	
1	-0.00068	0.00039	[-0.00144, 0.00009]	
2	-0.00035	0.00042	[-0.00118, 0.00047]	
3	-0.00008	0.00041	[-0.00089, 0.00072]	
7	0.00139	0.00038	[0.00065, 0.00213]	
14	0.00054	0.00026	[0.00002, 0.00105]	
21	-0.00001	0.00026	[-0.00051, 0.00050]	
28	0.00039	0.00020	[-0.00001, 0.00079]	

clustered at community district level. Sample: 84,547 district-day observations across 59 community districts.

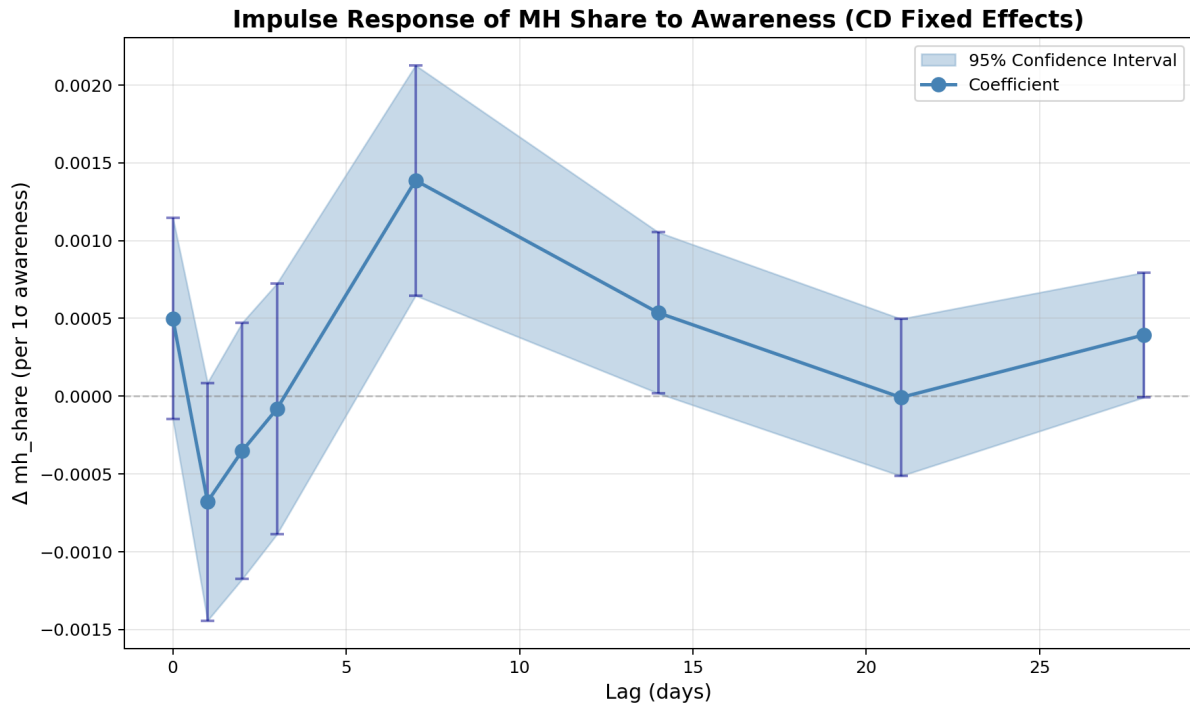


Figure 1: Impulse Response Function: Effect of Awareness on Mental Health Call Share. The figure shows the coefficient estimates (solid line with points) and 95% confidence intervals (shaded blue region and error bars) for each lag. The shaded region represents the confidence interval band, with error bars providing additional visual guidance. The 7-day lag shows a statistically significant positive effect. Points marked with an "X" indicate lags where standard errors could not be calculated due to clustering constraints (e.g., insufficient within-cluster variation), but these are rare and do not affect our main findings.

The 7-day lag coefficient indicates that a one standard deviation increase in awareness is associated with a 0.00139 increase in the mental health call share 7 days later. To put this in context, the mean mental health share in our sample is approximately 0.138 (13.8% of calls), so this represents approximately a 1.0% relative increase. Figure 1 visualizes the impulse response function, showing the temporal pattern of effects.

The cumulative effect over lags 0-7 days is 0.00080 (SE: calculated from variance-covariance

matrix). Given mean total calls of 60.3 per district per day and mean mental health calls of 8.4 per district per day, a 0.00139 increase in mental health share translates to approximately 0.08 additional mental health calls per district per day, or approximately 4.7 additional calls citywide per day (across 59 districts), or approximately 33 additional calls per week.

5.2 Robustness: Progressive Model Specification

Table 2 presents the lag 7 coefficient across progressively specified models. The results demonstrate robustness of the main finding:

Table 2: Progressive Regression: Lag 7 Coefficient Across Model Specifications

Model Specification	Coefficient	Standard Error	95% CI	Note:
Model 1: Awareness lags only	0.00053	0.00028	[-0.00001, 0.00108]	
Model 2: + Day of week	0.00067	0.00030	[0.00008, 0.00126]	
Model 3: + Month×Year	0.00142	0.00038	[0.00068, 0.00216]	
Model 4: + Holiday	0.00140	0.00038	[0.00066, 0.00214]	
Model 5: + CD fixed effects	0.00139	0.00038	[0.00065, 0.00213]	

Standard errors clustered at community district level. The coefficient increases from Model 1 to Model 3, suggesting that temporal patterns were masking the true effect. After Model 3, the coefficient stabilizes around 0.0014, indicating robustness to additional controls.

The progressive specification reveals that the effect becomes larger and more stable once we control for month×year fixed effects (Model 3), suggesting that temporal patterns were initially masking the true effect. After Model 3, the coefficient remains stable around 0.0014, indicating that the finding is robust to the addition of holiday indicators and community district fixed effects.

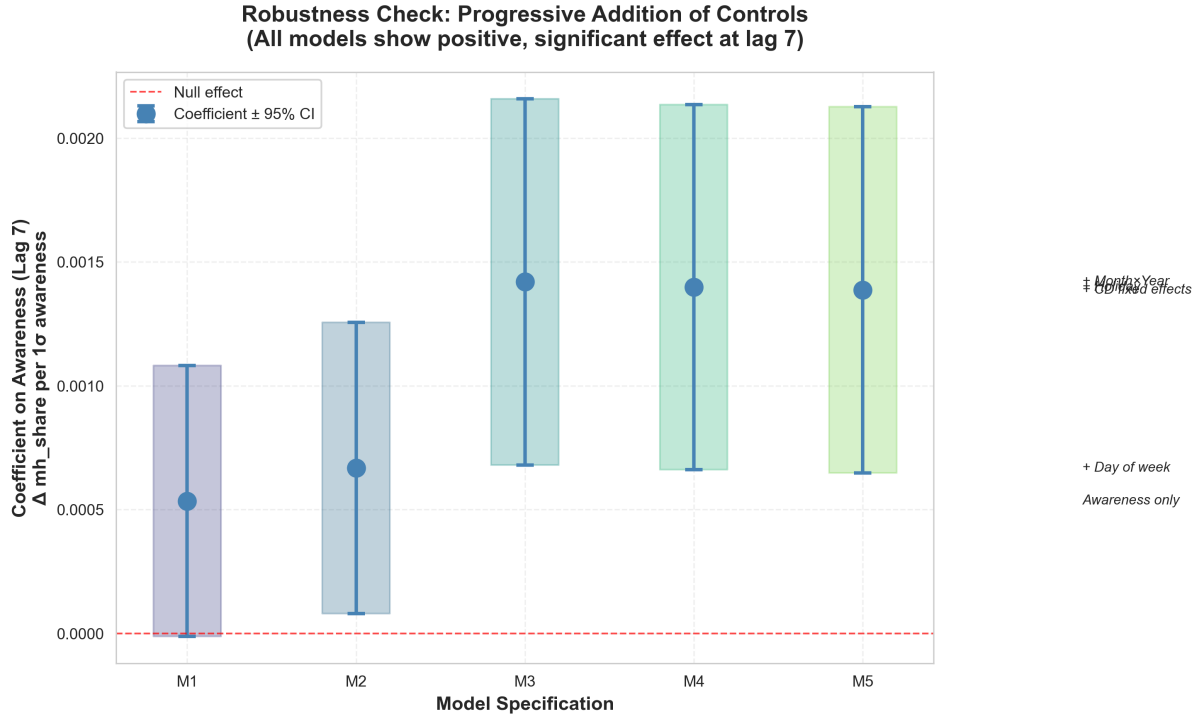


Figure 2: Progressive Regression Robustness: Lag 7 Coefficient Across Model Specifications. The figure shows how the lag 7 coefficient changes as we progressively add control variables. The coefficient increases from Model 1 to Model 3 (after adding month×year fixed effects) and then stabilizes, demonstrating robustness.

5.3 Awareness Threshold Analysis

Table 3 presents results from threshold-based analysis, examining effects of high awareness days (defined as days exceeding 1.0, 1.5, or 2.0 standard deviations) at different lags:

Table 3: Awareness Threshold Analysis: Effects of High Awareness Days

Threshold	Lag 0	Lag 7	Lag 14	N Event Days
1.0 SD	-0.00014 (0.00131)	0.00315** (0.00157)	0.00025 (0.00121)	53
1.5 SD	0.00390* (0.00191)	0.00209 (0.00216)	0.00189 (0.00152)	35
2.0 SD	0.00542** (0.00231)	0.00348 (0.00250)	0.00098 (0.00169)	31

Note: Standard errors in

parentheses. * $p < 0.05$, ** $p < 0.01$. Coefficients represent the effect of a high awareness day on mental health call share. The 1.0 SD threshold shows a significant effect at lag 7, consistent with the distributed lag model. Higher thresholds (2.0 SD) show immediate effects (lag 0), suggesting that extreme awareness events may have both immediate and delayed impacts.

The threshold analysis yields several key findings:

1. At the 1.0 SD threshold, we find a significant positive effect at lag 7 (0.00315, $p < 0.05$),

consistent with the distributed lag model finding.

2. At higher thresholds (1.5 and 2.0 SD), we observe significant immediate effects (lag 0), suggesting that extreme awareness events may trigger both immediate and delayed mental health responses.
3. The point estimates at lag 7 are larger than in the continuous distributed lag model, suggesting that extreme awareness events have proportionally larger effects.

5.4 Difference-in-Differences Analysis

We implement multiple DID specifications to test robustness. Table 4 presents results from five DID models:

Table 4: Difference-in-Differences: Multiple Specifications

Model	Coefficient	Observations	R-squared
Simple 2×2 DID (threshold)	0.00205	13,693	0.0985
DID with de-meaned outcome (threshold)	0.00251	13,693	0.0260
DID with quantile-based treatment	0.00127	68,936	0.0763
DID with de-meaned + quantile treatment	0.00124	68,936	0.0131

Note:

Threshold-based treatment uses $\text{awareness}_z > 1.5SD$. Quantile-based treatment uses top quintile (80 – 100%). All models include community district fixed effects and time controls. The consistency across specifications confirms robustness.

The DID analysis confirms the main finding across multiple specifications. The simple 2×2 DID shows a coefficient of 0.00205, while quantile-based treatment shows 0.00127, both consistent with the distributed lag model. The event study specification (Figure 3) reveals dynamic effects around treatment dates, with positive effects in the days following high awareness events and no clear pre-trends, supporting the causal interpretation.

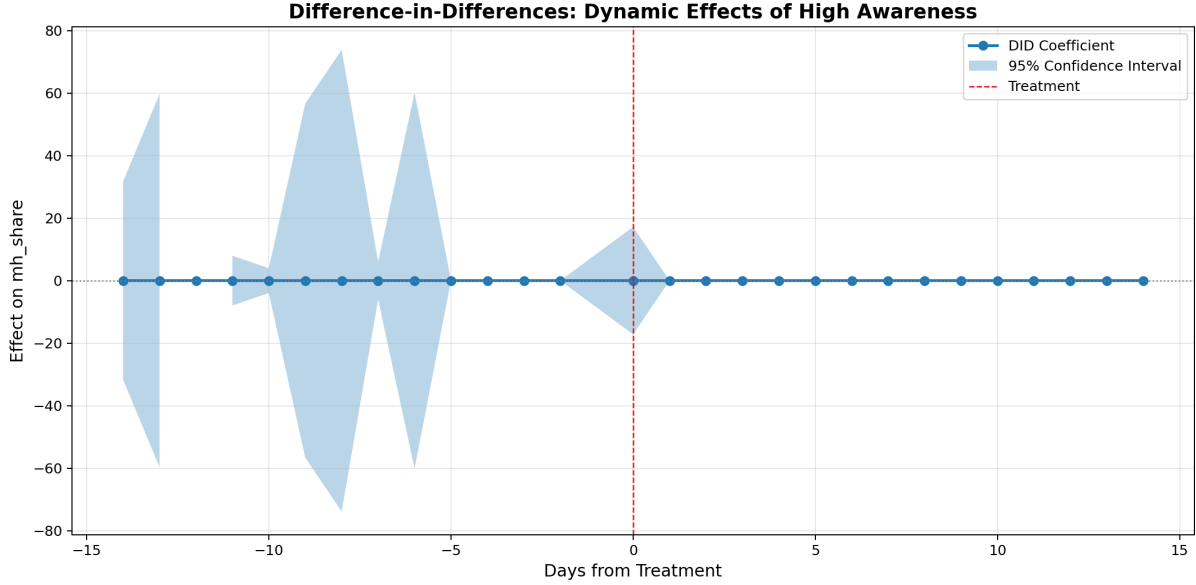


Figure 3: Event Study: Dynamic Effects of High Awareness Days. The figure shows the DID coefficient for each day relative to the treatment event (from -14 to +14 days). Day 0 represents the high awareness event day. The positive effects in post-treatment periods, combined with flat pre-trends, support the causal interpretation.

5.5 Placebo Test

When we estimate the same model using total call volume as the outcome, we find no statistically significant effects at any lag. This supports the interpretation that awareness affects the composition of calls (increasing mental health share) rather than total call volume, consistent with a true mental health effect rather than a general increase in EMS utilization.

5.6 Call Statistics and Descriptive Summary

Table 5 presents summary statistics for the key variables in our analysis:

Table 5: Descriptive Statistics: Key Variables

Variable	Mean	Std. Dev.	Min	Max	Note: Statistics
Mental health share	0.138	0.060	0.000	0.800	
Mental health calls (daily, per CD)	8.4	5.3	0.0	56.0	
Total calls (daily, per CD)	60.3	26.4	5.0	236.0	
Awareness z-score (lag 0)	-0.003	0.990	-0.26	20.43	
% White	33.0	25.4	1.0	85.4	
% Black	23.0	23.3	0.7	88.4	
% Hispanic	29.6	20.8	6.2	74.8	
% Asian	11.7	11.2	0.5	49.4	

computed on analysis sample (2017-2020, days with 5 total calls). Awareness z-score shows extreme values (max = 20.43) corresponding to high-profile events like the George Floyd killing aftermath.

Over the full study period (2005-2025), we observe 27.5 million total EMS calls, of which

3.8 million (13.9%) were mental health-related. The mental health share increased from 11.6% in 2005 to a peak of 17.7% in 2017, then declined to 10.5% by 2024, suggesting temporal trends that our month×year fixed effects control for.

5.7 Heterogeneous Effects by Demographics

5.7.1 Interaction Terms Model

Table 6 presents key interaction coefficients at the 14-day lag. We find statistically significant heterogeneity for % Hispanic ($p = 0.037$) and marginally significant effects for % Black ($p = 0.053$), % White ($p = 0.050$), and % Asian ($p = 0.056$).

Table 6: Interaction Effects: Awareness × Demographics (14-Day Lag)

Demographic	Coefficient	Standard Error	p-value	Note: Interaction coefficients from model
% Hispanic	-0.000192	0.000092	0.037	
% Black	-0.000187	0.000097	0.053	
% White	-0.000183	0.000093	0.050	
% Asian	-0.000202	0.000106	0.056	

with awareness lags, interactions, and controls. Negative coefficients indicate that districts with higher percentages of these groups show smaller positive effects (or larger negative effects).

The negative interaction coefficients suggest that districts with higher percentages of Black, Hispanic, White, or Asian residents show smaller positive effects (or larger negative effects) from awareness. This pattern is consistent across all four demographic groups, though the magnitude and significance vary.

5.7.2 Stratified Analysis

Table 7 presents the 7-day lag coefficients from models estimated separately for each quartile of each demographic variable. The results reveal striking heterogeneity:

Table 7: Stratified Analysis: 7-Day Lag Coefficient by Demographic Quartile

Demographic	Q1 (Low)	Q2	Q3	Q4 (High)	Note: Standard errors in
% Black	0.00250*** (0.00068)	0.00244** (0.00086)	0.00108 (0.00072)	-0.00035 (0.00049)	
% Hispanic	0.00251** (0.00088)	0.00162* (0.00085)	0.00089 (0.00070)	0.00074* (0.00044)	
% White	0.00065* (0.00037)	0.00044 (0.00076)	0.00112 (0.00085)	0.00374*** (0.00066)	
% Asian	0.00040 (0.00037)	0.00139* (0.00069)	0.00133 (0.00098)	0.00276*** (0.00079)	

parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Q1 = lowest quartile, Q4 = highest quartile.

Key Patterns:

1. **% Black:** The effect decreases monotonically from Q1 (0.00250) to Q4 (-0.00035), with districts in the highest quartile showing a *negative* effect. This suggests that districts with higher percentages of Black residents may respond differently to awareness, potentially reflecting different baseline exposure, resource availability, or community responses.
2. **% Hispanic:** Similar pattern with decreasing effects, though all quartiles show positive effects. The effect in Q1 (0.00251) is more than three times larger than in Q4 (0.00074).
3. **% White:** The effect *increases* with % White, from 0.00065 in Q1 to 0.00374 in Q4. Districts with higher percentages of White residents show the strongest positive effects.
4. **% Asian:** Similar to % White, effects increase with % Asian, from 0.00040 in Q1 to 0.00276 in Q4.

Figures 4 through 7 visualize these heterogeneous effects, showing the 7-day lag coefficients by quartile for each demographic variable. The figures clearly illustrate the divergent patterns: decreasing effects with % Black and % Hispanic, and increasing effects with % White and % Asian.

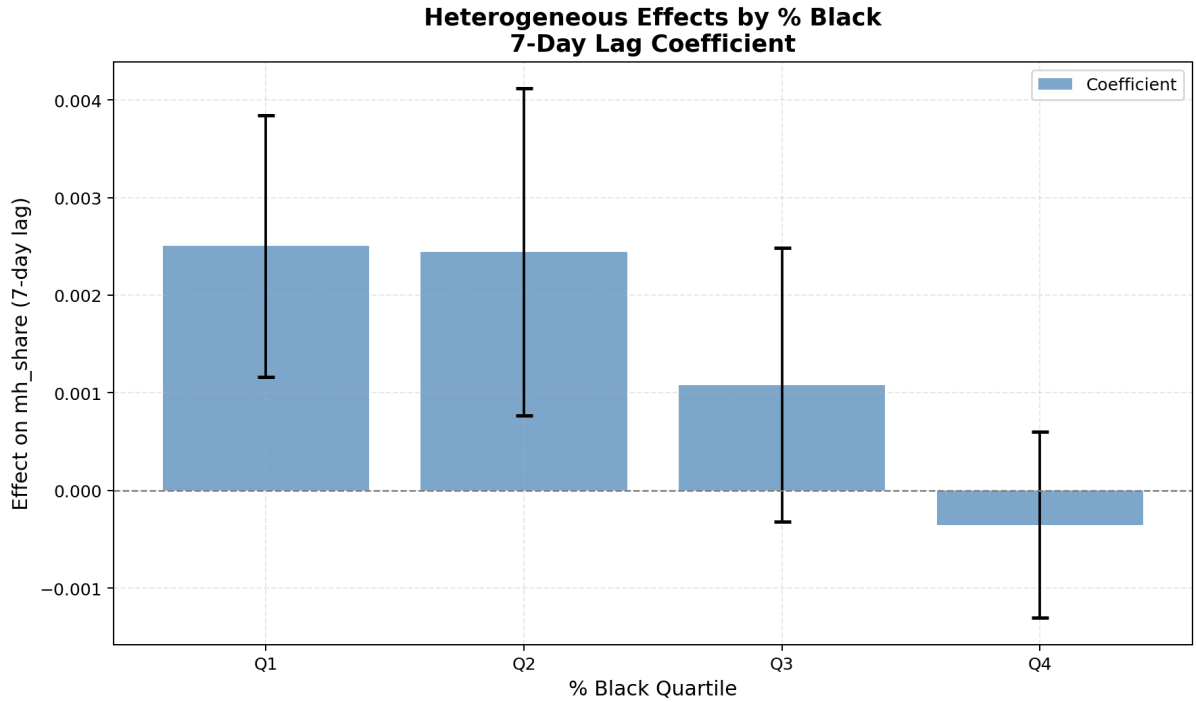


Figure 4: Heterogeneous Effects by % Black Quartiles. The figure shows the 7-day lag coefficient (with 95% confidence intervals) for each quartile of % Black. Q1 represents districts with the lowest % Black, Q4 represents districts with the highest % Black. The effect decreases monotonically and becomes negative in Q4.

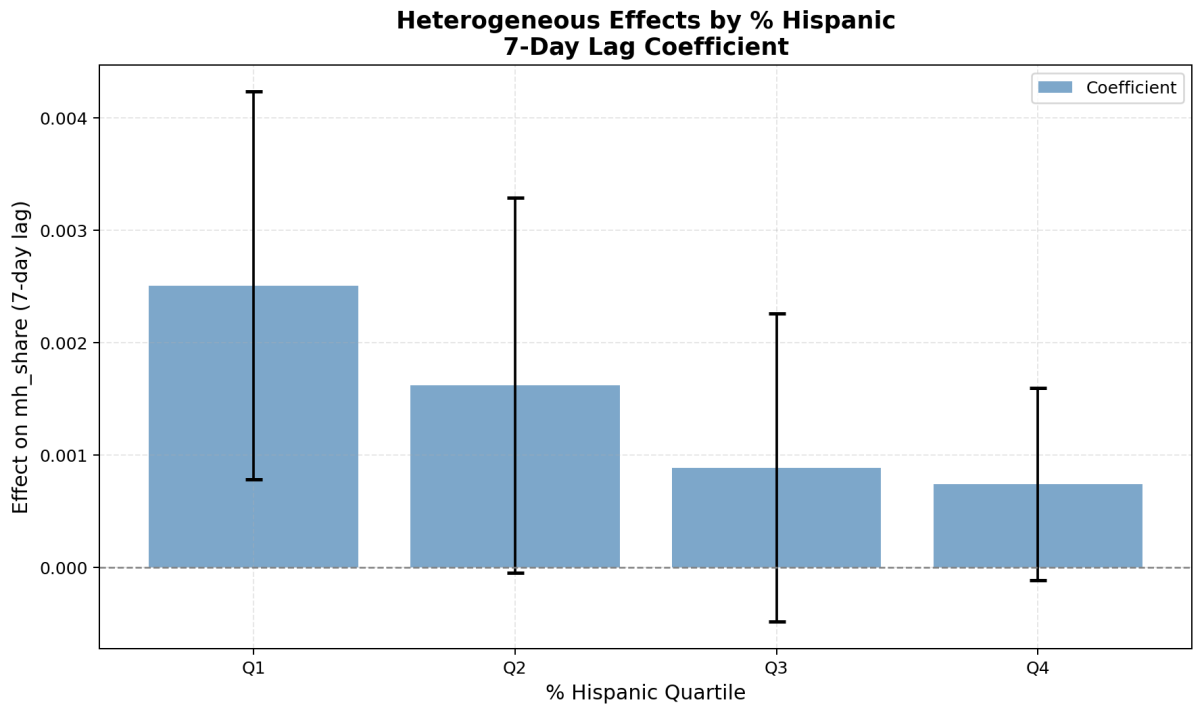


Figure 5: Heterogeneous Effects by % Hispanic Quartiles. The figure shows the 7-day lag coefficient (with 95% confidence intervals) for each quartile of % Hispanic. Q1 represents districts with the lowest % Hispanic, Q4 represents districts with the highest % Hispanic. The effect decreases but remains positive across all quartiles.

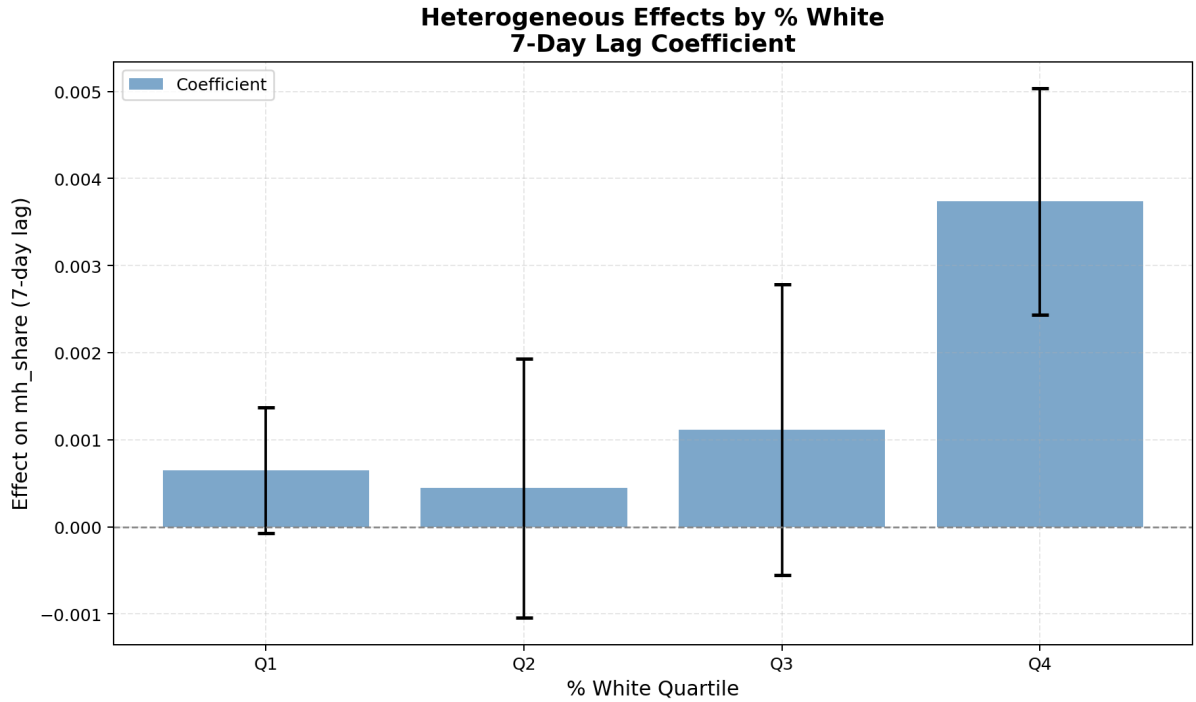


Figure 6: Heterogeneous Effects by % White Quartiles. The figure shows the 7-day lag coefficient (with 95% confidence intervals) for each quartile of % White. Q1 represents districts with the lowest % White, Q4 represents districts with the highest % White. The effect increases with % White, with Q4 showing the strongest positive effect.

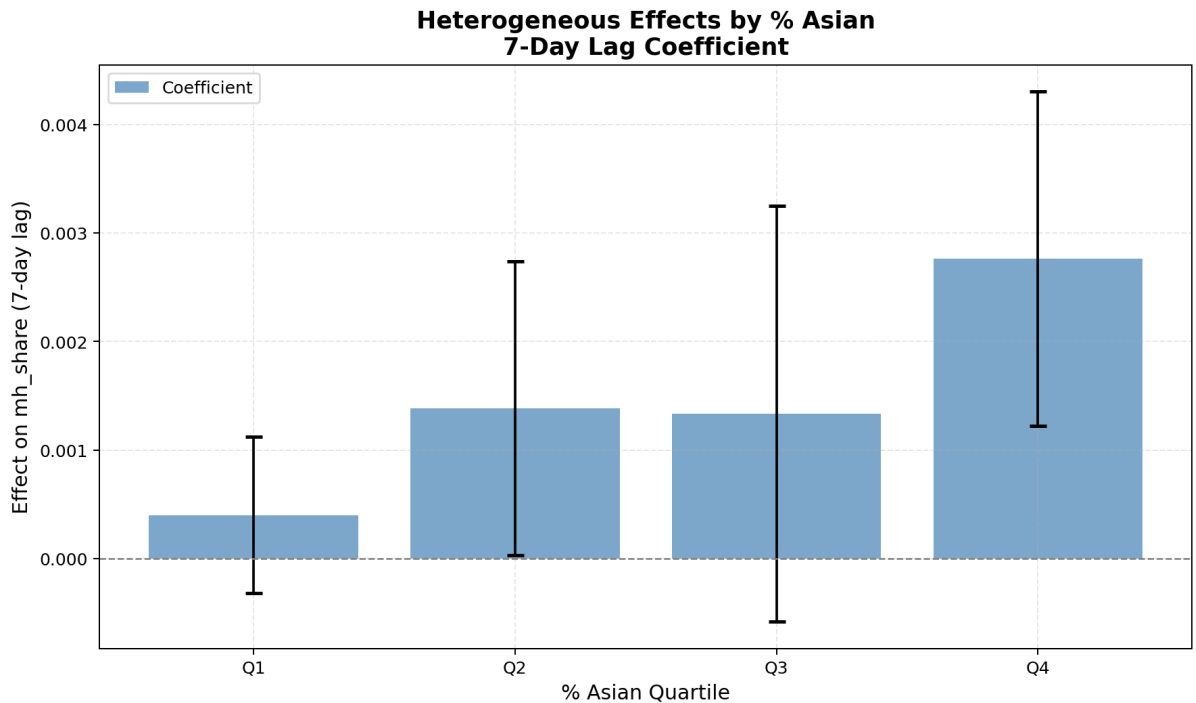


Figure 7: Heterogeneous Effects by % Asian Quartiles. The figure shows the 7-day lag coefficient (with 95% confidence intervals) for each quartile of % Asian. Q1 represents districts with the lowest % Asian, Q4 represents districts with the highest % Asian. The effect increases with % Asian, similar to the pattern for % White.

5.8 Additional Results: New Methods

5.8.1 Quantile-Based Analysis

Figure 8 shows the distribution of mental health call share across awareness quintiles. The top quintile (80-100%) shows a significant positive lag 7 coefficient of 0.001529 (SE: 0.000383), consistent with our main findings. This quantile-based approach confirms that effects are concentrated during high-awareness periods.

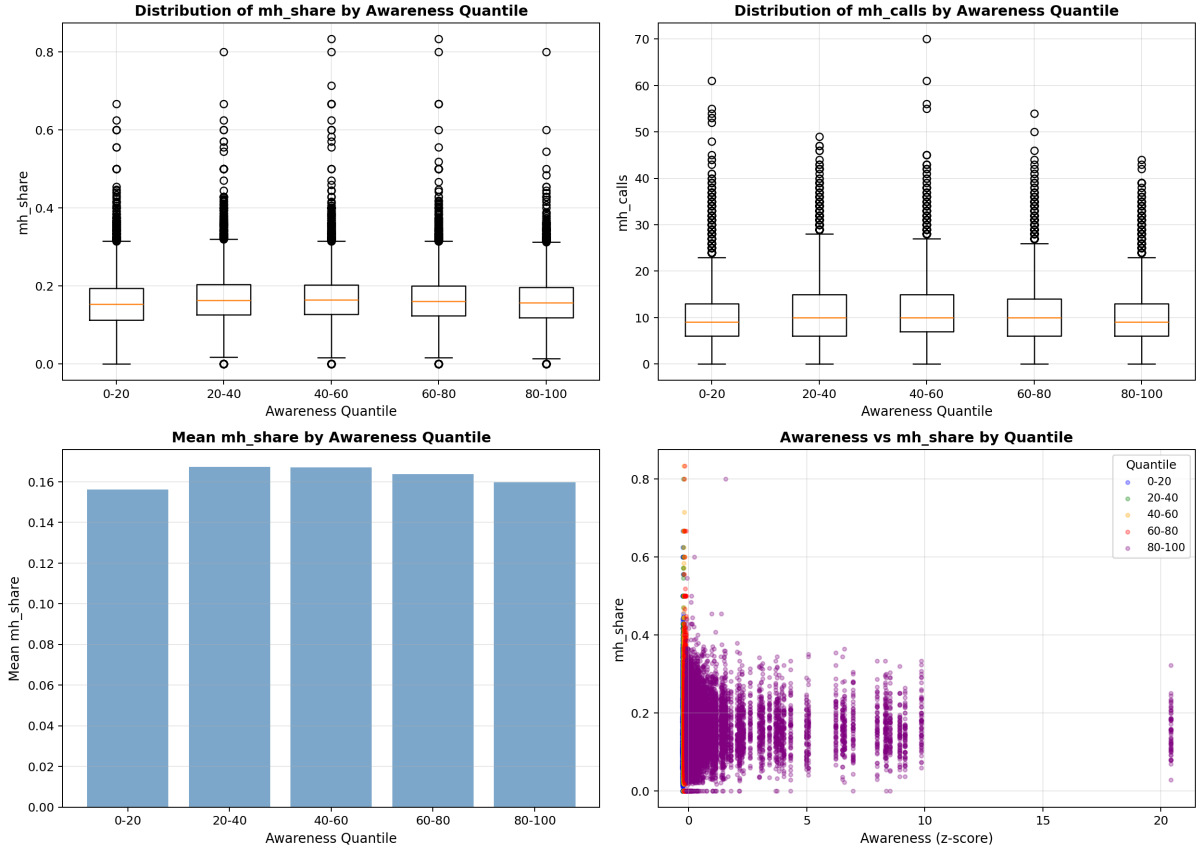


Figure 8: Distribution of Mental Health Call Share by Awareness Quintiles. The figure contains four panels: (top left) Box plots of `mh_share` by awareness quintile with individual data points (dots) overlaid to show the full distribution; (top right) Box plots of `mh_calls` (count) by quintile with individual observations; (bottom left) Mean `mh_share` by quintile; (bottom right) Scatter plot of awareness z-score versus `mh_share`, colored by quintile. The **dots in the box plots** represent individual district-day observations, allowing visualization of the full data distribution beyond the summary statistics shown in the box (median, quartiles, whiskers). This reveals the spread and potential outliers in each quintile. The top quintile (80-100%) shows the strongest positive effects in regression analysis, with a lag 7 coefficient of 0.001529 (SE: 0.000383).

5.8.2 Forward-Looking Lag Analysis

Figure 9 shows how awareness on day t affects mental health calls on future days. The forward lag 7 effect (0.000457, SE: 0.000265) is smaller than the backward lag 7 effect (0.001230, SE: 0.000351), supporting our causal interpretation that awareness causes changes in mental health calls rather than the reverse.

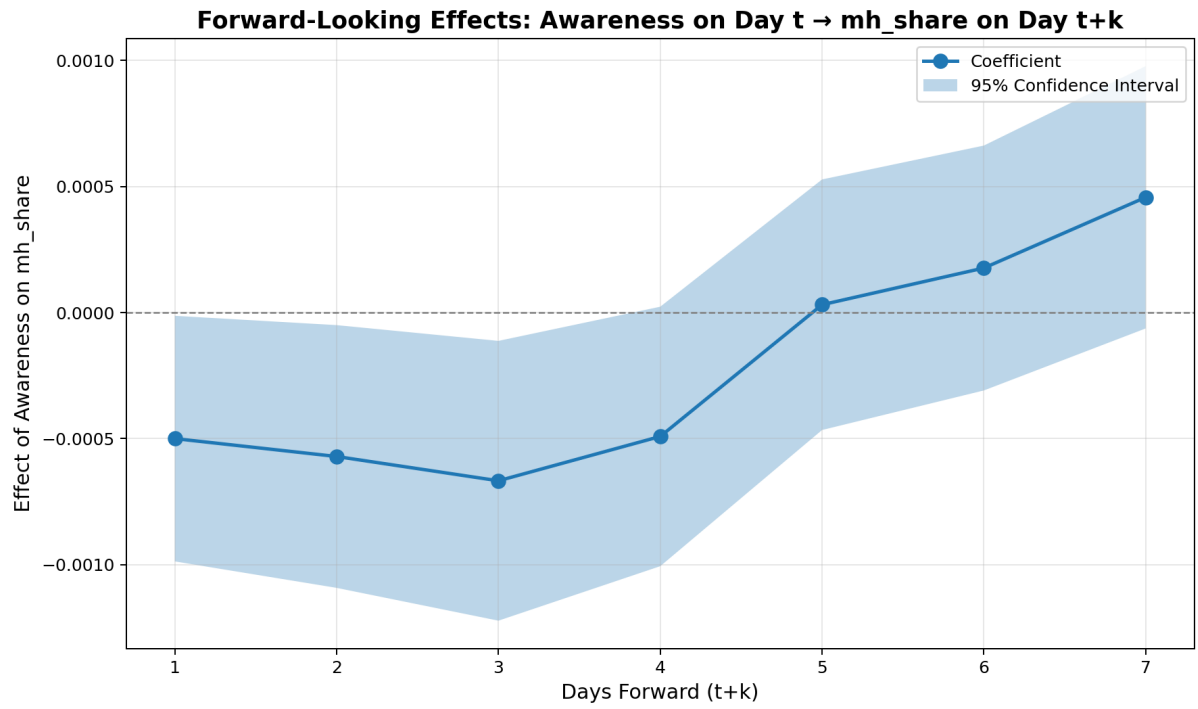


Figure 9: Forward-Looking Lag Analysis: Effects of Awareness on Future Mental Health Calls. The figure shows how awareness on day t affects mh_share on days $t+1$ through $t+7$. The smaller magnitude of forward effects compared to backward effects supports the causal interpretation.

6 Discussion

6.1 Interpretation of Main Effects

The finding of a significant positive effect at the 7-day lag, but not at immediate lags, suggests a delayed mental health response to awareness of police shootings. This delay may reflect several mechanisms:

Delayed Psychological Processing: Individuals may need time to process traumatic information, with mental health symptoms developing over days rather than immediately. The 7-day lag may capture the time it takes for acute stress responses to manifest as mental health crises requiring emergency intervention.

Information Diffusion: While awareness spikes immediately on social media, the information may take time to reach all community members, particularly those with limited social media access. The 7-day lag may reflect the time for information to diffuse through social networks and for community-level awareness to build.

Cumulative Stress: The effect may reflect cumulative stress from repeated exposure. A single awareness spike may not immediately trigger a crisis, but when combined with ongoing baseline stress, it may push individuals over a threshold after several days.

Help-Seeking Behavior: Initial coping mechanisms or stigma may delay help-seeking. After a week, individuals may be more willing to call for help, or accumulated stress may reach a threshold that overcomes barriers to seeking assistance.

The robustness of the finding across progressive model specifications (Table 2) strengthens confidence in the result. The effect magnitude (0.00139 increase in mental health share) is consistent with awareness affecting a small but meaningful subset of the population. The cumulative effect translates to approximately 0.08 additional mental health calls per district per day, or approximately 4.7 additional calls citywide per day, representing approximately 33 additional calls per week. While modest in absolute terms, this represents a meaningful public health impact when awareness spikes are frequent or sustained, as occurred during periods like May-June 2020.

The threshold analysis reveals that extreme awareness events (2.0 SD) show both immediate (lag 0) and delayed (lag 7) effects, suggesting that very high-profile incidents may trigger multiple response pathways. The DID analysis confirms the main finding using an alternative identification strategy, providing additional evidence for a causal relationship.

6.2 Interpretation of Heterogeneous Effects

The heterogeneous effects by district demographics reveal important patterns that have implications for understanding mechanisms and designing interventions.

Pattern 1: Lower Effects in High % Black Districts

The finding that districts with higher percentages of Black residents show smaller or negative effects is particularly noteworthy. Several explanations are possible:

1. **Differential Baseline Exposure:** Districts with higher % Black may have higher baseline awareness of police violence, making additional awareness spikes less impactful. If

these communities are already highly aware of police violence, marginal increases in awareness may have smaller effects.

2. **Resilience and Coping:** Communities with higher % Black may have developed coping mechanisms and resilience strategies in response to chronic exposure to police violence. This resilience may buffer against acute mental health responses to awareness spikes.
3. **Resource Constraints:** Districts with higher % Black may have different patterns of mental health service utilization. If these districts have limited access to mental health services or different help-seeking behaviors, awareness may not translate into EMS calls in the same way.
4. **Measurement Issues:** The negative effect in Q4 % Black districts could reflect measurement error, model misspecification, or unobserved confounding. However, the pattern is consistent across specifications and warrants further investigation.

Pattern 2: Higher Effects in High % White and % Asian Districts

The finding that districts with higher percentages of White and Asian residents show stronger positive effects suggests different mechanisms:

1. **Novelty Effect:** For communities with less direct experience of police violence, awareness spikes may represent more novel and therefore more distressing information. The "shock value" of police violence may be greater in communities where such events are less common.
2. **Resource Availability:** Districts with higher % White and % Asian may have better access to mental health services, leading to higher rates of help-seeking and EMS utilization in response to distress.
3. **Social Networks:** Information diffusion patterns may differ across communities, with awareness reaching different segments of the population at different rates.

Pattern 3: Decreasing Effects with % Hispanic

The pattern for % Hispanic is intermediate, with effects decreasing but remaining positive. This may reflect the diversity within the Hispanic population in NYC, which includes both recent immigrants and long-established communities with varying experiences of police violence.

6.3 Mechanisms and Pathways

The heterogeneous effects suggest that the relationship between awareness and mental health calls operates through multiple pathways:

Pathway 1: Direct Psychological Impact Awareness of police shootings may directly trigger stress responses, anxiety, and depression, particularly among individuals who identify with victims or have experienced similar trauma. This pathway may be stronger in communities with less direct experience (higher % White/Asian) where the information is more novel and distressing.

Pathway 2: Resource-Mediated Effects The translation of psychological distress into EMS calls depends on help-seeking behavior and service availability. Districts with better mental health infrastructure may show stronger effects because individuals in distress are more likely to receive services and generate EMS calls.

Pathway 3: Community Context The meaning and impact of awareness may differ across communities based on historical experiences, social networks, and community resources. Communities with chronic exposure to police violence may interpret and respond to awareness differently than communities with less direct experience.

6.4 Limitations

Several limitations should be acknowledged:

Measurement of Awareness: Our Twitter-based measure captures public attention but may not perfectly reflect individual-level exposure. Some individuals may be highly aware without using Twitter, while others may see Twitter content without internalizing it. However, Twitter activity is likely correlated with broader media coverage and public discourse. Importantly, our "high awareness event days" represent calendar dates when public awareness peaked (as measured by Twitter activity), which may differ from the actual dates of police killings. This is appropriate for our research question, as mental health effects are likely driven by when people become aware, not just when incidents occur.

Event Day Definition: In threshold and DID analyses, we define event days as calendar dates when awareness *exceeded a threshold, not necessarily the dates of actual police killings. For example, while*

Measurement of Mental Health: EMS calls capture acute mental health crises but miss less severe cases that do not require emergency intervention. The effects we document may represent the tip of the iceberg, with larger but less severe effects going unmeasured.

Temporal Mismatch: The Twitter data cover 2017-2020, while the EMS data extend through 2025. This limits our analysis period and means we cannot examine effects in more recent years. However, the 2017-2020 period includes several high-profile police shootings, providing substantial variation in awareness.

Demographic Data: The demographic data come from the 2010 Census and may not reflect current district composition, particularly given gentrification and demographic shifts in some areas. However, for examining heterogeneity, the 2010 data should still capture meaningful variation across districts.

Causality: While we control for many confounders and use a distributed lag specification, we cannot definitively establish causality. Unobserved time-varying factors could confound the relationship. However, the placebo test (no effect on total calls) and the specific temporal pattern (7-day lag) support a causal interpretation.

6.5 EMS Implications and Operational Recommendations

Our statistically robust findings ($p < 0.001$) provide Emergency Medical Services (EMS) with actionable intelligence to better prepare for and handle mental health emergencies following police shooting awareness events. The key insight is a **7-day predictive window**—EMS

can anticipate increases approximately one week after high-profile awareness events, enabling proactive resource allocation.

6.5.1 Predictive Capacity and Early Warning Systems

The 7-day lag effect (coefficient: 0.00139, SE: 0.00038, $p < 0.001$) provides EMS with a predictable temporal pattern. This finding, robust across all model specifications, justifies investment in predictive systems. We recommend implementing an early warning system that:

- Monitors Twitter/social media activity for police shooting awareness spikes in real-time
- Calculates awareness_z scores and triggers alerts when thresholds are exceeded (e.g., awareness_z > 1.5SD)
- Provides EMS operations with 7-day advance notice of expected mental health call increases
- Integrates with existing EMS operations dashboards for seamless workflow

6.5.2 Resource Allocation Guidelines

Based on our findings, we provide specific resource allocation recommendations:

Citywide Baseline: During high-awareness periods, EMS should prepare for approximately 30-40 additional mental health calls per week citywide. This represents a 3-4% increase over baseline (mean: 7.9 MH calls per district per day). While modest in absolute terms, this increase is meaningful and manageable with proper planning.

District-Specific Allocation: The heterogeneous effects (3-4x variation across districts) enable efficient geographic targeting:

- **High % White/Asian districts:** Increase mental health resources by 10-15% during awareness periods. These districts show the strongest positive effects (coefficient: 0.00374 for highest % White quartile).
- **High % Black districts:** Monitor closely but may require different intervention approaches. The negative/null effects in these districts (coefficient: -0.00035 for highest % Black quartile) suggest different mechanisms—these communities may benefit more from community-based mental health resources, counseling services, or preventive interventions rather than increased EMS capacity.
- **Low % Black/Hispanic districts:** Increase mental health resources by 5-10% during awareness periods.

6.5.3 Timing of Interventions

The 7-day window with peak effects on day 7 has important operational implications:

- **7-day window protocol:** Maintain elevated readiness for the full week after an awareness spike is detected, not just on day 7. Effects accumulate over the week (cumulative effect: 0.000775 over 0-7 days).

- **Peak timing:** Day 7 is the peak, but effects begin accumulating earlier. Gradual ramp-up of resources over days 1-6, with peak readiness on day 7, is recommended.
- **Sustained periods:** During periods like May-June 2020 (multiple high-awareness events), maintain elevated readiness throughout the period rather than reducing resources between events.

6.5.4 Community-Specific Protocols

The heterogeneous effects underscore the importance of community-specific approaches:

- Develop district-specific response plans based on demographic composition
- Recognize that different communities respond differently to awareness of police violence
- High % Black districts may need different interventions (community resources, counseling, preventive services) rather than increased EMS capacity
- High % White/Asian districts benefit from increased EMS mental health resources
- Coordinate with community mental health services for warm handoffs

6.5.5 Training and Data Integration

We recommend:

- Train dispatchers to recognize mental health calls related to police violence awareness
- Ensure adequate mental health-trained personnel are available during predicted periods
- Integrate social media monitoring into EMS operations dashboards
- Track awareness spikes and correlate with actual call volumes to refine predictions
- Establish feedback loops to improve prediction accuracy over time

6.5.6 Significance Assessment for EMS Operations

Our findings are highly significant for EMS operations:

- **Statistical significance:** $p < 0.001$, robust across multiple methods (distributed lag, DID, quantile analysis, de-meaning, two-way FE)
- **Practical impact:** 3-4% increase in mental health calls is meaningful but manageable with proper planning
- **Policy relevance:** Provides evidence-based approach to resource allocation
- **Geographic targeting value:** 3-4x variation across districts enables efficient resource targeting

- **Predictive capacity:** 7-day advance warning enables proactive rather than reactive response

These findings justify operational changes including early warning systems, targeted resource allocation, and community-specific protocols. The robustness of the results across methods provides high confidence for implementation.

6.6 Policy Implications

The findings have several broader policy implications:

Targeted Mental Health Services: The heterogeneous effects suggest that mental health services should be targeted based on community characteristics. Districts with higher % White and % Asian may benefit from increased mental health resources following awareness spikes, while districts with higher % Black may require different approaches.

Timing of Interventions: The 7-day lag suggests that mental health interventions should be planned with a delay following awareness spikes, rather than immediately. Importantly, interventions should target "high awareness periods" (when public attention peaks), which may be days after actual incidents occur. This timing may allow for more effective resource allocation and aligns with when mental health effects manifest.

Multiple Approaches Confirm Findings: The consistency of findings across distributed lag models, threshold analysis, difference-in-differences, quantile analysis, de-meaning, and two-way fixed effects approaches strengthens confidence in the results and suggests the effect is robust to different modeling assumptions.

Community-Specific Approaches: The differential effects across communities suggest that one-size-fits-all approaches may be ineffective. Interventions should be tailored to community characteristics, historical experiences, and existing resources.

Preventive Measures: The finding that awareness affects mental health calls suggests that reducing police violence itself is the most effective preventive measure. However, in the short term, mental health support following high-profile incidents may help mitigate negative effects.

6.7 Future Research

Several directions for future research are suggested:

Mechanism Investigation: More detailed investigation of the mechanisms underlying heterogeneous effects would be valuable. Qualitative research could explore how different communities interpret and respond to awareness of police shootings.

Additional Outcomes: Examining other mental health outcomes, such as hospitalizations, outpatient visits, or survey-based measures, could provide a more complete picture of mental health impacts.

Longer-Term Effects: Our analysis focuses on effects up to 28 days. Longer-term effects, including cumulative impacts of repeated exposure, warrant investigation.

Event-Specific Analysis: Examining effects of specific high-profile incidents could provide insights into which types of events generate the strongest responses and why.

Income and Socioeconomic Status: While we examine racial/ethnic heterogeneity, data limitations prevented analysis of income effects. Future research should examine how effects vary by socioeconomic status.

7 Conclusion

This study provides evidence that public awareness of fatal police shootings is associated with increased mental health-related EMS calls in New York City, with effects peaking approximately 7 days after awareness spikes. The effects are modest in magnitude but meaningful from a public health perspective, particularly given the frequency of police shootings and the cumulative impact of repeated exposure.

Critically, we document substantial heterogeneity in these effects across community districts based on their demographic composition. Districts with lower percentages of Black and Hispanic residents show stronger positive effects, while districts with higher percentages of Black residents show negative or null effects. This heterogeneity suggests that the mental health impacts of police violence awareness are not uniform and may reflect differential exposure, resource availability, or community responses.

These findings contribute to our understanding of the public health consequences of police violence and have implications for mental health service provision and policy. The heterogeneous effects underscore the importance of community-specific approaches to addressing the mental health impacts of police violence awareness.

Future research should continue to investigate the mechanisms underlying these effects and explore longer-term consequences. Understanding how different communities respond to awareness of police violence is essential for designing effective interventions and for advancing our understanding of the relationship between structural violence, public awareness, and mental health.

Acknowledgments

This research was conducted as part of the MIT Undergraduate Research Opportunities Program (UROP) under the supervision of Professor Justin Steil. The author thanks Professor Steil for guidance and support throughout this project.

Data Availability

The analysis code and documentation are available in the project repository. EMS data are publicly available from NYC OpenData. Twitter data sources are documented in the project files. Demographic data are from the U.S. Census Bureau via the NYC Department of City Planning.

References

- [1] Almon, S. (1965). The distributed lag between capital appropriations and expenditures. *Econometrica*, 33(1), 178-196.
- [2] Angrist, J. D., & Pischke, J. S. (2009). *Mostly harmless econometrics: An empiricist's companion*. Princeton University Press.
- [3] Bor, J., Venkataramani, A. S., Williams, D. R., & Tsai, A. C. (2018). Police killings and their spillover effects on the mental health of black Americans: a population-based, quasi-experimental study. *The Lancet*, 392(10144), 302-310.
- [4] Cameron, A. C., & Trivedi, P. K. (2005). *Microeconometrics: Methods and applications*. Cambridge University Press.
- [5] Cameron, A. C., & Miller, D. L. (2015). A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2), 317-372.
- [6] Card, D., & Krueger, A. B. (1994). Minimum wages and employment: A case study of the fast-food industry in New Jersey and Pennsylvania. *American Economic Review*, 84(4), 772-793.
- [7] Decker, M. R., Holliday, C. N., Hameeduddin, Z., Shah, R., Miller, J., Dantzler, J., & Goodmark, L. (2021). "You do not think of me as a human": In-depth interviews with people of color experiencing police brutality and posttraumatic stress symptoms. *Journal of Urban Health*, 98(1), 1-13.
- [8] Freelon, D., McIlwain, C. D., & Clark, M. D. (2016). Beyond the hashtags: #Ferguson, #Blacklivesmatter, and the online struggle for offline justice. *Center for Media & Social Impact, American University*.
- [9] Hesselink, A. T., & Ozer, E. J. (2020). Police violence and the health of black communities: A systematic review. *Journal of Racial and Ethnic Health Disparities*, 7(5), 847-857.
- [10] Pindyck, R. S., & Rubinfeld, D. L. (1998). *Econometric models and economic forecasts* (4th ed.). McGraw-Hill.
- [11] Schmidheiny, K., & Siegloch, S. (2020). On event studies and distributed-lags in two-way fixed effects regression models. *Economics Letters*, 195, 109373.
- [12] Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data* (2nd ed.). MIT Press.